

How Much Language Model Does a Speech Neuroprosthesis Need?

The stronger the decoder, the less rescoring helps

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Abstract

A speech neuroprosthesis decodes attempted speech from brain activity, and every high-performance system ends by rescoring the decoder’s candidate sentences with a language model. That stage is the only one that needs a multi-billion-parameter model, so it sets the hardware a patient has to carry. The published evidence for it comes from systems whose first-pass decoder was weak. Whether it still pays once the decoder improves has not been measured.

We hold the published T15 pipeline fixed, including its Kaldi decoder and OpenWebText 3-gram, and score one cached 100-candidate list per sentence with every rescorer, so the only thing that varies is the rescorer. At the published acoustic weight, sweeping 82M to 7.6B parameters moves word error from 8.1% to 7.8%, and after Holm correction only one arm, a closed model, separates from no rescoring. Re-tuning the decoder’s weight together with the rescorer lifts the 7B-class open models (OPT-6.7b to 7.2%, Qwen2.5-7B to 7.4%, both significant); nothing under 1B parameters helps under either protocol. Varying only the acoustic weight on fixed candidate lists, the gain from rescoring falls from 6.8 points when the decoder errs at 19.7% to 0.3–0.9 points at 8.1%. Asking a general language model to name the best candidate in a single pass fails: it answers from where a label sits in the list, not from the sentence beside it. The one purpose-built typed model we tested does not collapse and is the strongest arm, but its size is undisclosed. Every arm is untuned and text-only, so these are lower bounds.

1 Introduction

A speech neuroprosthesis decodes attempted speech from intracortical activity. A recurrent network trained with CTC [4] emits phoneme posteriors; a lexicon- and n -gram-constrained decoder turns those into a ranked list of candidate sentences; and a final stage rescoring that list with a large language model before the winner is spoken. In Willett et al. [1] that final stage takes word error from 23.8% to 17.4%. At that level of first-pass error, rescoring is unambiguously worth doing. The Brain-to-Text benchmark’s reference implementation pairs its decoder with OPT-6.7b [3]. Our question is whether that verdict survives the pipeline’s own improvement: as the first pass gets stronger, how much is there left for the second pass to fix?

We study one stage: text-only rescoring of a fixed N -best list, on the published T15 pipeline, whose decoder alone reaches 8.1% word error. Lattice rescoring, acoustic-conditioned reranking, generative correction and earlier fusion are different stages, and this design does not test them. The stage we do study is the only part of the pipeline needing a multi-billion-parameter model, so it sets the hardware a patient carries. We ask two questions the literature has largely settled by assumption:

1. **How large does the rescoring model have to be?** We sweep model scale across 82M–7.6B parameters at fixed task and fixed candidate lists.
2. **Does the stage have to be N forward passes?** The rescorer’s job is to pick one of N written-down options. We test whether posing it as a single constrained decision, a *typed* selection returning a candidate label, preserves accuracy, including under the mitigations proposed for listwise LLM reranking.

Our answers, measured on real neural data rather than simulation, are that once the decoder is strong the stage is worth under a point, that only 7B-class open models capture even that, and that the single-pass shortcut fails for general language models for a specific reason. Both are lower bounds: every arm is untuned and text-only, so neither result speaks to what a trained or acoustically conditioned rescorer could do. §2 covers where the literature finds larger gains.

Contributions.

- A scale study of the rescoring stage on real T15 recordings across a $93\times$ parameter range, holding the published pipeline’s candidate lists fixed, with bootstrap intervals and paired significance tests at each arm’s tuned interpolation weight, both at the published acoustic weight and with that weight re-tuned per arm (§4).
- A controlled sweep of first-pass strength. Varying only the decoder’s acoustic scale on fixed candidate sets, the rescoring gain falls as the first pass improves, with a second data point from an earlier, weaker pipeline (§4).
- A test of the combination rule: four interpolation schemes, a jointly fit multi-arm reranker, and MWER fine-tuning of the smallest rescorer leave the ranking unchanged (§5). The decoder’s own acoustic weight, which none of them touches, is what moves the 7B models (§4).
- A negative result on listwise typed selection with general language models. The measured cause is *label-position collapse*: models answer from where a label sits in the list rather than from the candidate beside it. Ablations over menu size and permutation aggregation separate two failure modes without repairing either (§6).
- An abstention analysis against a fair baseline, a temperature-calibrated softmax over per-candidate scores. Both shapes carry signal; the typed arm’s curve is the stronger (§7).

2 Related work

Rescoring. Second-pass rescoring is conventionally likelihood-based: interpolate the rescorer’s sequence likelihood with the first-pass score. Recent work reframes it discriminatively: Rescore-BERT [5] fine-tunes an encoder with a minimum-word-error (MWER) objective and a pooled sentence score, and Shivakumar et al. [9] extend discriminative training to pretrained GPT-2 and BERT rescors. We do not fine-tune any rescorer, so our curve measures what untuned models achieve. §5 tests how far untuned combination and discriminatively fit interpolation reach on fixed lists. A further step conditions the rescorer on acoustics rather than text: joint audio/text training scores candidates against first-pass audio embeddings [14], and speech-text foundation models extend this to large pretrained models [12]. Our arms are text-only by design.

Where the literature finds larger gains. Three settings report nontrivial rescoring gains our design does not cover, and each sharpens rather than contradicts our claim. *Discriminative training:* RescoreBERT [5] cuts relative WER by 6.6%/3.4% on LibriSpeech clean/other over a non-discriminative baseline, LoRB [8] matches it while training 0.08% of the parameters, and MWER fine-tuning helps both GPT-2 and BERT rescors [9]. We ran the direct test on our own lists and it does not reclaim the missing headroom (§5). *Stronger first passes with context:* Udagawa et al. [10] gain from bidirectional rescoring with in-domain fine-tuning and past/future context on a Conformer-Transducer baseline, larger where the first pass leaves more headroom or the rescorer sees more than one sentence. *Generative correction:* Yang et al. [11] surpass the N -best oracle by generating corrected output rather than selecting. Our typed arms are constrained to the menu by design, a safety property for a device that speaks for someone, so our negative result applies to fixed-list selection and not to open-ended correction.

Position bias in listwise LLM ranking. Our negative result is a domain-specific instance of a documented phenomenon. Tang et al. [6] show that listwise rankers are sensitive to the order in which candidates are presented, and propose *permutation self-consistency*: shuffle the list several times and aggregate the resulting orderings into a central ranking. Bito et al. [7] analyse this instability directly and argue that reducing exposure skew alone does not establish ranking validity. Architectural routes remove the order dependence structurally rather than by aggregation: InvariRank blocks cross-candidate attention and shares positional framing so that a single forward pass is order-invariant [15]. We test the Tang et al. mitigation in §6 and find that it discriminates between two distinct failure modes, repairing one and destroying the other, which to our knowledge has not been reported. Architectural invariance remains untested here: it requires model surgery or retraining we did not perform.

Long-form rescoring. We score whole sentences independently. Long-form ASR fuses neural LMs over wider context, and the gains grow with it [13]. Our task is per-sentence by construction: each list corresponds to one attempted sentence in a copy task with no conversational context to carry. Whether language-model help returns once the context window widens is the natural next test.

3 Setup

This paper proposes no method, so it has no Method section. It measures an existing pipeline stage, and this section fixes the instantiation: the data, the candidate lists every arm scores, the rule that combines a rescorer’s output with the decoder’s, and the statistics. Results follow in §4–§7, each answering one of the questions above.

Neural data and acoustic model. We use the T15 copy-task recordings released with the Brain-to-Text benchmark [2]: 256 electrodes over speech motor cortex, 512 features binned at 20 ms. Phoneme posteriors come from the authors’ own pretrained GRU decoder (5 layers of 768 units), run unmodified. Evaluation uses the `val` split; the `test` split is released without ground-truth sentences, so word error is not computable on it, and the reference implementation likewise reports on `val`. We refer to the 419/978 partitions as development/test throughout, although both originate from the benchmark’s `val` split.

Candidate lists. We use the published pipeline end to end: a Kaldi/WFST decoder with an n -gram trained on OpenWebText produces 100-best lists, combined as $0.3 \cdot \text{acoustic} + n\text{-gram}$. This gives 1397 lists with oracle WER 2.5% and the truth present in 89% of lists. Lists are cached, so every arm scores byte-identical candidates, and we compare rescorsers *to each other on identical lists* rather than to published absolute numbers. (An earlier revision used a custom flashlight decoder with 60-best lists and a weaker first pass at 14.1% WER; those numbers return only as a second data point in §4.)

Score combination. Each rescorer returns one score per candidate, combined with the decoder’s score as $s(h) = (1 - \alpha)\ell(h) + \alpha r(h)$, both z -normalised across the list. For typed arms, $r(h)$ is the log of the model’s probability for that candidate’s label, floored at 10^{-6} ; for per-candidate arms it is the length-normalised sequence log-likelihood. We call each rescorer under test an *arm*. Per-candidate arms are the GPT-2 family [16] (124M–774M parameters), DistilGPT-2 (82M), OPT-6.7b [3] and Qwen2.5-7B; typed arms are OPT-6.7b, Qwen2.5-7B, Laya (421M) and the hosted Jev model. α is tuned *per arm* on a 419-sentence development split and applied to the disjoint 978-sentence test split.

Robustness of the combination rule. The tuned weight jitters without changing the outcome: across 200 bootstrap resamples of the development split α spreads over roughly 10 grid values per arm (std 0.10–0.16) while test WER moves 0.10–0.21pp, because the WER surface is flat near the optimum. Normalising per list matters more. A single global mean and standard deviation costs every arm 0.08–0.42 points and leads development tuning to discard the language model altogether (retuned $\alpha \leq 0.05$ on four of six arms), since one scale cannot align log-likelihoods across lists of different lengths.

Statistics. Word error is corpus-level word-level Levenshtein computed with jiwer 4.0.0 on the raw strings (already lowercase and punctuation-free), with no additional normalisation; the bootstrap code that produced every interval and p -value is released with the data. Intervals are percentile bootstrap over sentences (2,000 resamples). Arm-vs-arm comparisons use a paired bootstrap (5,000) *on the test split only*, with each arm at its own tuned α , so the test matches what the table reports.

Cost. We report parameters and FLOPs per decision ($\approx 2 \times \text{params} \times \text{tokens read}$) rather than wall-clock, which confounds the model with the accelerator and, for hosted models, the network. Measured token counts are in Appendix B: a per-candidate arm reads 343 tokens per decision and a typed arm 191–213, so the two shapes read comparable amounts of text and the compute difference is dominated by model size, not by the number of forward passes.

4 How large does the rescorer need to be?

At the published acoustic weight, rescoring is barely worth doing. The first pass alone scores 8.1% WER and the best open rescorer reaches 7.8%. Table 1 and Figure 1 sweep 82M to 7.6B parameters, and the per-candidate curve is flat to within measurement noise. **DistilGPT-2 (82M) lands exactly on the baseline at 8.1%**, and the full 93 $\times$ range buys three-tenths of a point. The spread is smaller than the uncertainty on any single arm: every 95% interval in Table 1 spans about two points and every interval overlaps every other, so the marginal

Rescorer	Shape	WER	95% CI	Sent. acc.	Params	TFLOP
<i>Per-candidate scoring</i>						
OPT-6.7b		7.8	[6.8, 8.8]	71.6	6700M	1.49
Qwen2.5-7B		7.8	[6.8, 8.7]	70.9	7600M	1.69
GPT-2 large		7.8	[6.9, 8.8]	71.3	774M	0.17
GPT-2 medium		7.9	[6.9, 8.9]	70.7	355M	0.08
GPT-2 small		8.0	[7.0, 9.0]	70.8	124M	0.03
DistilGPT-2		8.1	[7.1, 9.1]	69.5	82M	0.02
<i>Typed selection (one pass over the whole list)</i>						
Jev (hosted)		7.5	[6.5, 8.4]	71.3	—	—
Laya		8.1	[7.1, 9.1]	69.9	421M	0.13
OPT-6.7b		8.1	[7.1, 9.1]	69.9	6700M	2.56
Qwen2.5-7B		8.1	[7.1, 9.1]	70.0	7600M	3.24
No rescoring (first pass only)	—	8.1	[7.1, 9.1]	69.9	—	—
<i>Oracle (best in list)</i>	—	2.5	—	—	—	—

Table 1: Real T15 data, 978 held-out test sentences, identical 100-best lists from the published decoder, α tuned per arm on a disjoint development split. Bold: the efficiency comparison. At the published acoustic weight (0.3), across a $93\times$ parameter range word error moves three-tenths of a point; after Holm correction only Jev clears significance. Table 3 re-tunes that weight.

estimates alone cannot separate a 82M rescorer from a 7.6B one. The paired bootstrap below is the sensitive test, because it holds the sentence sample fixed across arms, and it does not separate them either.

Table 2 gives explicit significance, and it disciplines the reading. Against no rescoring, Jev (-0.63 , $p = 0.006$) is the only arm that survives Holm correction for the seven arms tested (adjusted $p = 0.042$); OPT-6.7b (-0.29 , raw $p = 0.035$) does not (adjusted $p = 0.21$), and neither do Qwen ($p = 0.064$, adjusted 0.256), GPT-2 large ($p = 0.050$, adjusted 0.25), or any smaller model (all adjusted $p \geq 0.25$). Within open models, scale buys nothing measurable at this weight: OPT-6.7b against GPT-2 small is -0.18 at $p = 0.10$. That reading depends on holding the decoder’s acoustic weight at its published value, which was chosen for the decoder alone.

Re-tuning the decoder’s weight. Once a rescorer is added, the acoustic weight is one more setting of the combined system, and nothing ties it to 0.3. We therefore tune it jointly with α on the development split (acoustic weight over 17 values in $[0, 5]$, α over 21 in $[0, 1]$), evaluate once on the test split, and compare against no rescoring, whose own development-tuned weight is again 0.3 (Table 3). The two 7B-class open models now separate clearly: OPT-6.7b reaches 7.2% (-0.89 , Holm-adjusted $p < 0.001$) and Qwen2.5-7B 7.4% (-0.67 , adjusted $p = 0.010$), while no model under 1B parameters does (all adjusted $p \geq 0.27$). Scale now buys something: OPT-6.7b beats GPT-2 small by 0.70 points ($p < 0.001$). The optimum is broad rather than a lucky grid point: every setting within 0.05 of it in both weights lands between 7.20% and 7.40% on test. The closed typed arm is best at 6.9%, with its tuned acoustic weight at 2.0, where it largely takes over the n -gram’s role, but it does not separate from OPT-6.7b (-0.28 , 95% CI $[-0.72, +0.15]$, $p = 0.11$). The revised reading is that the stage is worth under a point once the decoder is strong, that only 7B-class models capture it, and that the published fixed-weight protocol understates it by about two-thirds.

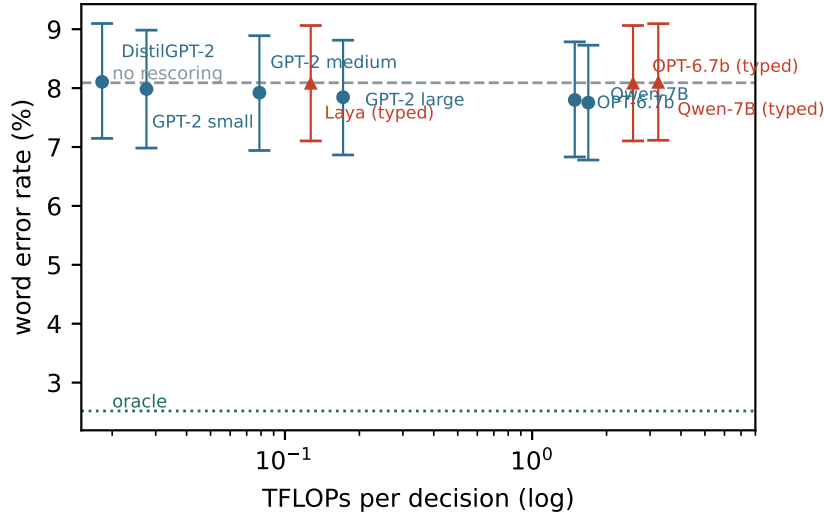


Figure 1: Word error against compute per decision (log scale). Circles are per-candidate scoring, triangles typed selection, all at the published acoustic weight. The curve is flat across two orders of magnitude of compute; re-tuning that weight (Table 3) tilts it in favour of the 7B models. Plotted points are the exact Table 1 values; the table rounds to one decimal.

The gain depends on first-pass strength. The same arms on our earlier weaker first pass (14.1% WER, oracle 7.3%, truth in list 67%) told the opposite story: OPT-6.7b reached 10.9%, a 3.2-point gain, and DistilGPT-2 at 12.9% captured about a third of it. Those two pipelines differ in decoder, language model and menu size, so on their own they are an observed association rather than a controlled comparison. We therefore vary first-pass strength directly: holding the decoder and the candidate sets fixed, we sweep the acoustic scale from 0 to 1.0 and re-rank. This moves first-pass error from 19.7% down to 8.1% at the published scale of 0.3, degrading again past it, with a single variable changed. It is a rerank of fixed candidates, not a new beam decode.

Figure 2 shows the result for OPT-6.7b, with α re-tuned on the development split at each scale. The gain is largest when the first pass is weak (6.8 points at 19.7% WER) and shrinks to 0.3 points at 8.1%. Past the published scale the gain grows again, partly because the first pass degrades and partly because a heavier acoustic weight lets the rescorer take over the n -gram’s role: at 0.35 the first pass is unchanged at 8.09% while OPT-6.7b reaches 7.43%, and the joint optimum above adds 0.9 points. Over the range a practitioner would operate in, the value of rescoring is a decreasing function of first-pass strength, measured with one variable changed. This is the paper’s central result.

Error analysis: substitutions, and when overrides are right. Errors are dominated by substitutions: per 100 reference words the first pass makes 7.52 substitutions, 0.21 insertions and 0.35 deletions; OPT-6.7b moves those to 7.20/0.29/0.31 and Jev to 6.63/0.41/0.41. Rescoring trims substitutions and pays for part of it in insertions, but the bulk of the error mass is untouched.

On the 94 test sentences where OPT-6.7b overrides the decoder’s top hypothesis, we relate the outcome to the decoder’s own top-1/top-2 score margin. Overrides of a confident decoder are the most likely to be right, 53% correct in the high-margin tercile against 26% and 16% in

Comparison (A vs. B)	Δ WER	95% CI	p	p_{Holm}
GPT-2 large (per-cand.) vs. OPT-6.7b	+0.05	[-0.15, +0.24]	0.701	—
Qwen2.5-7B (per-cand.) vs. OPT-6.7b	-0.05	[-0.40, +0.30]	0.413	—
Jev (hosted) (typed) vs. OPT-6.7b	-0.34	[-0.80, +0.12]	0.082	—
Jev (hosted) (typed) vs. GPT-2 large	-0.38	[-0.85, +0.08]	0.057	—
Laya (typed) vs. GPT-2 large	+0.25	[-0.03, +0.53]	0.963	—
OPT-6.7b (typed) vs. OPT-6.7b	+0.29	[-0.02, +0.59]	0.972	—
Qwen2.5-7B (typed) vs. Qwen2.5-7B	+0.35	[-0.08, +0.77]	0.949	—
Jev (hosted) (typed) vs. No rescoring (first pass only)	-0.63	[-1.12, -0.14]	0.006	0.042
Qwen2.5-7B (per-cand.) vs. No rescoring (first pass only)	-0.34	[-0.76, +0.09]	0.064	0.256
OPT-6.7b (per-cand.) vs. No rescoring (first pass only)	-0.29	[-0.59, +0.02]	0.035	0.21
GPT-2 large (per-cand.) vs. No rescoring (first pass only)	-0.25	[-0.53, +0.03]	0.050	0.25
GPT-2 medium (per-cand.) vs. No rescoring (first pass only)	-0.17	[-0.40, +0.06]	0.079	0.256
GPT-2 small (per-cand.) vs. No rescoring (first pass only)	-0.11	[-0.48, +0.25]	0.315	0.63
DistilGPT-2 (per-cand.) vs. No rescoring (first pass only)	+0.02	[-0.23, +0.26]	0.587	0.63
OPT-6.7b (per-cand.) vs. GPT-2 small	-0.18	[-0.48, +0.11]	0.104	—

Table 2: Paired bootstrap on the test split, each arm at its tuned α and the published acoustic weight. p_{Holm} is the Holm step-down correction over the seven arms-vs-baseline comparisons; only Jev survives it.

the low and middle, but with point-biserial $r = 0.12$ the relationship is weak and non-monotonic, so margin does not reliably predict an override’s success. The predictable part of the gain sits exactly where the truth is in the list: overrides cannot reach the truth on the 11% of lists where it is absent.

5 The interpolation rule is not the bottleneck

The efficiency curve could in principle be an artefact of how we combine scores. We test that on the same cached lists with four variants, each tuned per arm on the development split and evaluated on the disjoint test split: **V0**, the paper’s rule (z -normalise both scores per list, combine with weight α); **V1**, raw length-normalised log-likelihood with a learned global scale; **V2**, per-arm temperature scaling with no extra weight; and **V3**, a discriminative MWER-style fit of the two scores on the development split. V3 is a two-parameter interpolation, not a trained reranker: no encoder is fine-tuned, so it is not comparable to RescoreBERT [5].

No variant changes the picture (Table 4). All four sit within noise on every arm, and none beats V0 significantly once the 18 arm-variant comparisons are accounted for (smallest raw $p = 0.019$, Holm-adjusted 0.33). Temperature scaling actively hurts the two smallest arms (GPT-2 small $7.98 \rightarrow 8.37$, DistilGPT-2 $8.11 \rightarrow 8.50$), a reminder that dev-fitted combination can overfit on 419 sentences. The ranking of arms is invariant to the rule. None of these variants, however, touches the decoder’s own acoustic weight, and §4 shows that weight matters more than any of them.

A joint linear reranker over the first-pass score and five per-candidate arms, fit with the same objective, reaches 7.68% against 7.8% for the best single arm (paired $p = 0.161$), keeping OPT-6.7b and Qwen2.5-7B and zeroing the rest. It takes the first-pass score as one fixed feature, so it cannot re-weight acoustic against n -gram evidence; the joint re-tuning in §4 can, and gains more.

Arm	acoustic	α	WER	Δ vs. none [95% CI]	p	p_{Holm}
No rescoring	0.30	—	8.09	—	—	—
DistilGPT-2 (82M)	0.35	0.35	8.15	+0.06 [-0.31, +0.43]	0.651	0.651
GPT-2 small (124M)	0.40	0.40	7.91	-0.18 [-0.61, +0.26]	0.222	0.444
GPT-2 medium (355M)	0.45	0.40	7.75	-0.34 [-0.76, +0.10]	0.067	0.268
GPT-2 large (774M)	0.50	0.45	7.77	-0.32 [-0.83, +0.18]	0.114	0.343
OPT-6.7b	0.40	0.45	7.20	-0.89 [-1.35, -0.44]	<0.001	<0.001
Qwen2.5-7B	0.45	0.45	7.42	-0.67 [-1.15, -0.21]	0.002	0.010
Jev (hosted, typed)	2.0	0.75	6.93	-1.17 [-1.67, -0.68]	<0.001	<0.001
Laya (typed)	0.30	0.00	8.09	0.00	—	—
OPT-6.7b (typed)	0.25	0.20	8.37	+0.28 [+0.11, +0.47]	—	—
Qwen2.5-7B (typed)	0.30	0.20	8.11	+0.02 [-0.22, +0.25]	—	—

Table 3: The decoder’s acoustic weight re-tuned jointly with α per arm on the development split; test split, paired bootstrap (5,000, one-sided) against no rescoring; p_{Holm} over the same seven arms as Table 2. Bold: arms that survive correction. Open typed arms (bottom) remain at or below the baseline.

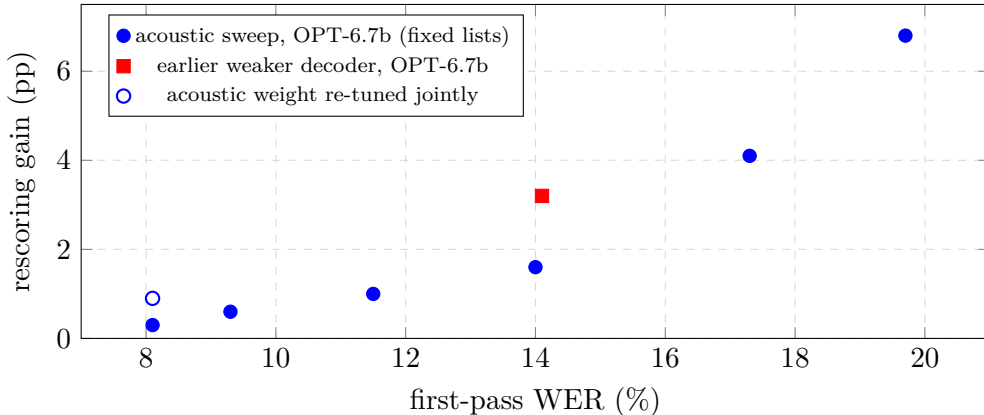


Figure 2: Rescoring gain collapses as first-pass strength increases. Blue points vary only the acoustic scale on fixed candidate sets; the red square is the earlier weaker decoder (a different pipeline, shown separately). Where the published decoder sits, at 8.1% error, three-tenths of a point remain at the published weight and 0.9 with that weight re-tuned (hollow circle).

Discriminative fine-tuning of the small rescorer. The fits above move scalars, not the model. The stronger test fine-tunes GPT-2 small itself with the MWER objective on the cached lists: each training list contributes its top-20 candidates, their length-normalised log-likelihoods become a posterior over the list, and the loss is the expected word-error count under that posterior, backpropagated through the full model. We train on 300 of the 419 development lists (119 held out for early stopping), AdamW at 10^{-5} , batches of 4 lists, 3 epochs on an L40S; the interpolation weight α is re-tuned on the development split and the final model is evaluated on the disjoint 978-sentence test split, which was used for neither training, early stopping, nor α selection. The pipeline reproduces both cached baselines exactly before training (frozen GPT-2 small at $\alpha = 0.4$: 8.0%; no rescoring: 8.1%). Held-out expected error falls from 0.201 to 0.195 across the three epochs, so the objective learns, but the test result is 8.2% WER at $\alpha = 0.5$, against 8.0% frozen and 8.1% for doing nothing (paired bootstrap, one-sided: $p = 0.62$ vs. no rescoring, $p = 0.84$ vs. frozen GPT-2 small). Three epochs of discriminative training on 300 lists

Rescorer	V0 (z-norm)	V1 (raw+scale)	V2 (temperature)	V3 (MWER-lin.)
OPT-6.7b	7.80	7.71	7.71	7.83
Qwen2.5-7B	7.75	7.77	7.60	7.77
GPT-2 large	7.85	8.09	8.11	7.88
GPT-2 medium	7.92	8.03	8.09	7.89
GPT-2 small	7.98	8.12	8.37	8.03
DistilGPT-2	8.11	8.06	8.50	8.03

Table 4: Test WER for four score-combination schemes on identical lists, each tuned per arm on the development split. No variant beats V0 once the 18 arm-variant comparisons are accounted for (paired bootstrap, one-sided; best raw $p = 0.019$ for DistilGPT-2 under V3, Holm-adjusted $p = 0.33$). V3 is a linear MWER-style fit of the two scores, not a trained reranker.

Typed rescorer	Labels used	Mass on A	Truth sel.	Truth sel. ($\neq A$)
Jev (hosted)	57 / 100	72%	67.9%	26.6%
Laya	44 / 100	69%	53.7%	2.3%
OPT-6.7b	13 / 100	35%	27.3%	0.2%
Qwen2.5-7B	31 / 100	28%	29.3%	2.4%
<i>Top-1, no rescorer</i>	—	—	68.0%	—

Table 5: Diagnostic for the typed arms over all 1397 lists (100-best generation; mean 38.7 candidates per list). “Labels used” counts distinct candidate labels ever selected; “truth selected” is how often the arm’s own argmax is correct, before combination. Label A is the decoder’s top candidate. The last column conditions on deviating from it: only Jev’s deviations carry signal.

do not separate the small rescorer from the baseline by a measurable margin. That is one recipe on one model: it shows this fine-tuning did not recover the headroom, not that discriminative fine-tuning cannot.

6 Listwise typed selection fails, and why

If the rescorer’s job is to pick one of N written-down options, an obvious efficiency move is to ask for exactly that. It does not work for general language models: the two open typed arms land on the no-rescoring baseline (Table 1), with OPT-6.7b and Laya both tuning to $\alpha = 0$, meaning the combination discards them entirely, and only the closed Jev arm improves on it. Re-tuning the acoustic weight does not rescue the open typed arms (Table 3), while Jev becomes the strongest arm overall.

Table 5 shows the mechanism, reproduced under the strong first pass. Candidates are serialised in the decoder’s ranking order, so label A is the decoder’s own top hypothesis, and all four typed arms collapse onto it, placing 28–72% of their decisions on that one label while using only 13–57 distinct labels across all lists. This is the domain-specific form of the positional prior documented by [6, 7]. What separates the arms is not the prior but the deviations from it. Conditional on picking any label other than A, Jev selects the truth 27% of the time; Laya, OPT-6.7b and Qwen2.5-7B manage 2%, 0% and 2%. Only Jev’s deviations carry signal. Three arms are therefore a strong prior over the first position plus noise; Jev is a strong prior over the first position plus informative corrections, which is why it alone beats taking the decoder’s top hypothesis (68.0% sentence accuracy) and doing nothing.

Menu size N	5	10	20	60
Truth available in menu	56.5%	60.5%	64.5%	66.8%
Qwen2.5-7B (typed)	76.1%	69.1%	59.2%	1.5%
OPT-6.7b (typed)	22.1%	40.0%	52.4%	55.1%
<i>Take first-pass top-1</i>	<i>80.5%</i>	<i>75.2%</i>	<i>70.5%</i>	<i>68.2%</i>
<i>Permutation mitigation at $N = 60$ (Borda over 5 shuffles)</i>				
Qwen2.5-7B	single order	1.5%	Borda-5	56.9%
OPT-6.7b	single order	55.1%	Borda-5	2.7%

Table 6: Accuracy *conditional on the truth being present in the menu*, which differs by N and must be divided out. The italic row is the do-nothing baseline: take the decoder’s top hypothesis. Subset of 400 sentences. No configuration beats the decoder’s top hypothesis; permutation self-consistency rescues Qwen while destroying OPT.

The prompt asks for the “most fluent, natural English sentence,” which could bias selections toward fluency over correctness. It does not appear to. Using the cached per-candidate GPT-2 log-likelihoods as a fluency proxy, Jev picks the most fluent candidate on 40.7% of test lists but the truth on 77.6%. Conditional on deviating from label A with the truth present, it picks the truth 40.2% of the time against 24.7% for the most fluent candidate. On the sharp subset where truth and fluency disagree and Jev deviates ($N = 144$), it picks the true-but-less-fluent candidate 31.9% of the time and the fluent-but-wrong one 19.4%, with 48.6% going to neither. Its deviations lean toward the true transcript *despite* the fluency wording, not because of it: though with $N = 144$ a mild fluency lean cannot be ruled out. Label A is the fluency argmax on only 35.6% of lists, so position and fluency are entangled at the top but not identified with each other.

6.1 Menu size and permutation aggregation

Two obvious objections are that the menu is too long and that the prompt order is arbitrary. We test both on the weaker-regime 60-best lists, where the typed arms’ failure was first characterised; the mechanism finding carries over to the published regime above. Because shorter menus contain the truth less often (from ... 56.5% at $N = 5$ to 66.8% at $N = 60$), raw selection accuracy is not comparable across N . Table 6 therefore reports accuracy conditional on the truth being available.

The two failures separate cleanly. **Qwen2.5-7B is menu-length sensitive:** it reaches 76.1% at $N = 5$, using every available label, and degrades monotonically to 1.5% at $N = 60$. **OPT-6.7b is not:** it uses three of five labels even at $N = 5$ and never approaches the baseline. Applying permutation self-consistency (Borda over five shuffles) at $N = 60$ separates them further: it *rescues* Qwen (1.5% $\rightarrow$ 56.9%, roughly its $N = 20$ level) and *destroys* OPT (55.1% $\rightarrow$ 2.7%). This is what one should expect: aggregation recovers signal that order noise obscured, but a model with a fixed positional prior has no candidate-dependent signal to aggregate, so shuffling converts its answer into noise. The mitigation therefore doubles as a diagnostic for which failure mode is present.

Neither repair is sufficient. Qwen’s best configuration (76.1% at $N = 5$) remains below the 80.5% obtained by taking the decoder’s top hypothesis, and its rescued $N = 60$ result (56.9%) remains below 68.2%. *No general language model we tested performs listwise typed selection well*

Arm	Truth in list (874 sent.)			Truth absent (104 sent.)		
	WER	Δ vs none	p	WER	Δ vs none	p
No rescoring	4.37	—	—	36.56	—	—
OPT-6.7b	4.05	-0.31	0.035	36.42	-0.13	0.433
Qwen2.5-7B	4.04	-0.33	0.077	36.16	-0.40	0.342
GPT-2 small	4.28	-0.09	0.346	36.29	-0.27	0.360
Jev (typed)	3.80	-0.57	0.015	35.50	-1.06	0.123

Table 7: Test WER split by whether the reference transcript is in the candidate list. Gains concentrate where the truth is present; where it is absent, no arm moves the needle. Paired bootstrap vs. no rescoring within each group.

Rescorer	Shape	ECE	sentence error among spoken, at coverage				
			30%	50%	70%	90%	100%
Jev (hosted)	typed	0.088	3%	8%	13%	22%	29%
Laya	typed	0.324	9%	10%	19%	28%	30%
OPT-6.7b	typed	0.220	9%	13%	21%	27%	30%
Qwen2.5-7B	typed	0.095	9%	16%	21%	25%	30%
OPT-6.7b	per-cand. + softmax	0.160	8%	10%	17%	24%	28%
GPT-2 large	per-cand. + softmax	0.136	9%	13%	18%	25%	29%
Qwen2.5-7B	per-cand. + softmax	0.116	9%	15%	18%	25%	29%

Table 8: Coverage-risk. Per-candidate arms receive a temperature-calibrated softmax over their candidate scores, temperature fit on the development split, so they are not disadvantaged merely by lacking a native probability.

enough to be worth running at all.

7 Where the gains live, and abstention

The reference transcript is absent from the candidate list on $100 - 89 = 11\%$ of sentences, and there no rescorer can succeed. Splitting the test set on that boundary (874 vs. 104 sentences) locates the entire effect: OPT-6.7b improves in-list WER by -0.31 points ($p = 0.035$) and Jev by -0.57 ($p = 0.015$), while out-of-list WER barely moves for any arm (Table 7).

This bounds what abstention can buy. Selecting the 50% most confident sentences *within each group*, sentence error is 100% whenever the truth is absent, for the typed arm and for the calibrated per-candidate baseline alike: there is nothing to select. Abstention helps only by declining sentences no rescorer could have fixed.

A fair comparison requires giving per-candidate arms a confidence, which they lack natively; we fit a softmax temperature on the development split. That baseline has teeth (Table 8): OPT falls from 28% error at full coverage to 8% at 30%. Jev’s curve is the strongest (29% to 3%), but the gap is one of degree, and of *ranking* rather than calibration. Isotonic recalibration introduces zero rank inversions and leaves the curve unchanged, and the max softmax probability already beats the top-1/top-2 margin among per-candidate signals. At full coverage OPT and Jev have the same sentence error (28.4% vs. 28.7%); Jev’s curve falls faster because its confidences separate right from wrong decisions more sharply (0.736 vs. 0.409, against OPT’s 0.647 vs. 0.368). This

is one closed model, not a property of typed decisions.

Candidate-set size. Pruning to the decoder’s top 20 raises the oracle from 2.5% to 3.3% and drops truth-in-list from 89% to 86%, bounding what any selection method could reach. The efficiency conclusion survives: with α retuned on pruned development lists, Qwen2.5-7B and GPT-2 large remain significant ($p = 0.024$, $p = 0.045$), OPT slips below the bar ($p = 0.12$), and the $93\times$ parameter spread is 0.4pp at $N = 20$ against 0.3pp at $N = 100$.

8 Limitations

- **One participant, one split, limited power.** T15, val, 978 test sentences. Effects here are under a point and bootstrap intervals span roughly a point, so the study detects effects around half a point and above. Read “no improvement” as “none large enough for this sample to see.” No other T15 session with released transcripts was available, so cross-session generalisation is untested.
- **Text-only, untuned, no earlier fusion.** No arm conditions on acoustics or is fused before the second pass, so the curve does not bound what acoustic access could achieve [14]. On discriminative fine-tuning we have one test (§5): one model, one objective, 300 training lists. A larger effort could still shift the curve.
- **Results depend on the protocol.** Holding the decoder’s acoustic weight at its published value, as our main tables do, understates the stage by about two-thirds; we report both protocols. The joint search over two weights is fit on 419 sentences.
- **The first-pass sweep reranks fixed candidate sets.** Varying the acoustic scale changes first-pass strength without re-running the beam decode, so the candidate pool is fixed by construction.
- **Typed failures are not exhaustive.** Four models, one prompt template (Appendix A), one label construction, one aggregation scheme. The instruction asks for the “most fluent, natural English sentence,” which may bias selection toward fluency over correctness; we quantify that confound in §6 but did not test alternative wordings. The menu-size and Borda ablations use the earlier 60-best lists; the collapse itself is confirmed on the published lists in Table 5.
- **One typed arm is closed,** so both the working listwise result and the abstention observation rest on a model a reader cannot inspect. We release its cached lists, prompt template, label mappings, per-sentence label probabilities and the α settings, so the analysis is reproducible even though the model is not.

9 Conclusion

On real intracortical recordings with the published decoder, the language-model rescoring stage contributes under a word-error point. At the published acoustic weight, a $93\times$ range of model size moves word error three-tenths of a point and no open model clears significance after Holm correction. Re-tuning that weight with the rescorer roughly triples the gain for the 7B-class open models ($8.1\% \rightarrow 7.2\%$ for OPT-6.7b), which then separate from both the baseline and

the small models; nothing under 1B parameters does under either protocol. The strongest arm under both is closed (Jev [17]) and does not separate from OPT-6.7b; we read it as evidence that a purpose-built typed model can do the job, not as a deployable or cost result.

Collapsing the stage into a single listwise typed decision does not work for general language models. Every typed arm piles most of its decisions on the decoder’s top candidate, and only Jev’s deviations from that prior carry signal. Permutation self-consistency rescues one failure mode while destroying the other, without making either competitive with doing nothing.

The stage mattered when the first pass was weaker. On our earlier 14.1%-error decoder it was worth 3.2 points, and a controlled sweep varying only the acoustic scale puts the gain at 6.8 points at 19.7% first-pass WER against 0.3–0.9 at 8.1%, depending on whether the acoustic weight is re-tuned. Four interpolation schemes, a jointly fit reranker and a discriminatively fine-tuned GPT-2 small all land within noise; the one setting that matters is the decoder’s own weight, and even with it re-tuned the gain at a strong first pass is a fraction of the gain at a weak one.

One hypothesis fits the remaining puzzle. Oracle WER is 2.5% against 8.1% first-pass, yet text-only models exploit almost none of that headroom. A plausible explanation is that the errors a strong first pass leaves are increasingly acoustically ambiguous rather than resolvable from candidate text alone. Our data cannot settle this, since every arm here is text-only; an acoustically conditioned rescorer on the same lists would.

A Prompt template, serialisation and label alphabet

Typed arms receive one prompt per list. Candidates are serialised one per line as `<label>: <sentence>`, in the decoder’s own ranking order unless stated otherwise, with the instruction block held fixed:

```
A speech decoder produced these candidate sentences for what a person was
trying to say. Exactly one is what they meant.
Answer with the single label of the most fluent, natural English sentence.
A: <candidate 1>
B: <candidate 2>
...
Answer:
```

For chat-tuned models the block is wrapped in the model’s own chat template and the assistant turn is opened with the scaffold "The answer is ", so that the first free token position coincides with the label. Labels are built greedily from A–Z, then decimal numerals, then a–z, then two-letter combinations AA–ZZ (the extension was needed for 100-option menus: some tokenizers yield only 61 single-token labels from the shorter pool), keeping a candidate only if it encodes to exactly one token in that model’s vocabulary and its id has not already been used; duplicates are skipped rather than reused, so the map from label to candidate is injective. Scores are read as the model’s log-probability at the single answer position, restricted to the label token ids and renormalised over the menu.

B Token accounting

Per-candidate arms read every candidate once: at a mean of 38.7 candidates per list and 32.2 characters each, 343 tokens per decision at ≈ 4 characters/token. Typed arms read the same

candidates plus labels and instructions in one prompt, a measured median of 191 tokens (OPT tokenizer) and 213 (Qwen). The two shapes read comparable amounts of text, so the compute difference in Table 1 follows model size, not the number of forward passes. FLOPs are $2 \times \text{params} \times \text{tokens}$, ignoring attention’s quadratic term, which is negligible at these lengths.

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